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Statistical Analysis of Risk and Returns on Varying Assets

The objective of this project is to examine the relationship between risk and short-term returns across financial assets. Specifically, I will investigate whether assets with higher market beta, a measure of systematic risk, tend to exhibit higher short-term returns compared to lower-beta assets. This is a common problem within financial theory, which suggests that investors should be expected to obtain higher returns for taking on greater risk. The experiment that I will be analyzing will be in the following format: the null hypothesis is that the mean short-term returns of high-beta stocks is less than or equal to that of low-beta stocks; the alternative hypothesis is that the mean short-term returns of high-beta stocks is greater than the mean short-term returns found in low-beta stocks

In addition to comparing average returns, this study also examines whether higher-beta stocks exhibit greater return volatility and higher downside risk. Downside risk is measured using maximum drawdown, which is the largest percentage decline from an initial price to the lowest observed price within a fixed time window. This metric reflects the potential for short-term losses, which is particularly relevant for investors using leveraged strategies or stop-loss rules. Also, in order to make the project slightly more interesting and diverse (I also wanted to personally explore this part myself), a similar framework is applied to cryptocurrencies! Since traditional market beta is not directly defined for crypto assets, Bitcoin (BTC) is used as a proxy for the market. This allows for the construction of a comparable “crypto beta” and enables a cross-asset comparison of risk-return relationships.

The dataset consists of approximately 30 publicly traded U.S. stocks from a range of sectors, along with a few major cryptocurrencies. Historical price data was collected using the Python library `yfinance`, which retrieves publicly available financial data from Yahoo Finance. Weekly adjusted closing prices over a six-month period were used to compute returns and risk measures. The code itself consisted of loading in the target assets via the `yfinance` library, cleaning the data, computing target variables needed for the dataset, and finally returning the data in the form of a csv file.

For each asset, the following variables were constructed:

- **Beta:** A measure of sensitivity to market movements. S&P 500 and BTC served as respective comparisons.
- **Average Weekly Return:** The mean of weekly percentage returns over the sample period.
- **Volatility:** The standard deviation of weekly returns, representing variability in performance.
- **Maximum Drawdown:** The largest percentage decline from a peak to a trough over the sample period, capturing downside risk.
- **Type:** Indicator distinguishing between stocks and cryptocurrencies.

A small subset of the dataset can be found below:

	Asset	Type	Avg_Return	Volatility	Max_Drawdown	Beta
1	AAPL	Stock	-0.000573441	0.033372742	-0.110489532	1.1993980734
2	ABBV	Stock	-0.001812978	0.0322035352	-0.125106005	-0.039815823
3	AMD	Stock	0.0136092741	0.096171465	-0.258972589	0.1383173612
4	AMZN	Stock	-0.001649472	0.0468545103	-0.196418509	1.7156290099
5	AVGO	Stock	-0.001412459	0.0555143582	-0.227952822	2.5271096813
6	BAC	Stock	-0.003069682	0.0340538007	-0.168864018	1.2792992329
7	BRK-B	Stock	-0.001349459	0.020879258	-0.067807159	0.3684796841
8	BTC-USD	Crypto	-0.018319604	0.0578810486	-0.467765758	1
9	COST	Stock	0.0029574363	0.0310070684	-0.086103327	0.3484781988
10	CVX	Stock	0.0104362875	0.0258787689	-0.070304698	0.3965705417
11	ETH-USD	Crypto	-0.02502852	0.0765323088	-0.571113315	1.3019932464
12	GOOG	Stock	0.0083696632	0.0385613547	-0.118837359	1.4628154729
13	GOOGL	Stock	0.008781008	0.0388366847	-0.116804769	1.4275083986
14	JNJ	Stock	0.0116933375	0.0281023323	-0.04753346	0.0452542496
15	JPM	Stock	-0.003065374	0.0313641316	-0.135093592	1.1604697993
16	KO	Stock	0.0060270829	0.0239300807	-0.077194914	0.3333148272
17	LLY	Stock	0.0104175542	0.0525492625	-0.15934193	0.5998639569
18	MA	Stock	-0.004669674	0.0286235875	-0.142524302	0.9321620278
19	META	Stock	-0.007734883	0.0487039433	-0.200469061	1.4277384253

To better understand the structure of the dataset, 3 diagnostic plots were generated using SAS. They can be found respectively on the next page.

The histogram of beta values indicates a broad range of risk levels across assets, with most observations concentrated between approximately 0.5 and 1.5, and a small number of higher-beta outliers. This suggests that the dataset captures both relatively stable and more volatile assets, providing sufficient variation for further analysis.

The distribution of average weekly returns appears slightly skewed and includes several extreme values. This is consistent with financial return data, which often deviates from normality due to market fluctuations and occasional large price movements. The presence of outliers highlights the importance of using appropriate statistical methods when analyzing returns.

Finally, the scatter plot of beta versus volatility reveals a positive relationship between these variables. Assets with higher beta values tend to exhibit greater volatility, although there is still noticeable dispersion in the data. This suggests that even though beta is associated with increased risk, it does not fully explain the variability in returns.

Overall, these exploratory findings provide initial insight into the structure of the data and validates the use of this dataset for this project. The graphs can be found below!

